Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… □ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. □ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. □ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief

Access and independence

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

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TA brief · continued

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): the amylase and pH core practical is examined in [QUAL] Biology; enzyme assays like it are routine in brewing, baking and clinical laboratories, and a Level 3 science or laboratory-technician route builds directly on it.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Risk-note controls cross-checked against publicly accessible [SYS] [QUAL] Enzymes and [QUAL] Iodine on 26 September 2026. Teacher/technician must authorise the local method, concentrations, quantities, PPE and disposal using current member guidance. The full member-only [QUAL] method and [SYS] were not accessible for this check.

Use minimum enzyme and iodine concentrations and quantities. Prevent enzyme powder aerosol; do not spray solutions. Follow eye-protection and solvent-specific iodine controls.

Illustrative or supplied data is reasoning practice only and is never logged as practical completion. Log only an observed, authorised physical contribution; the teacher decides whether it meets the relevant course requirement.

Prompt ladder

Prompt ladder for adults

1. Point to the relevant place / instruction.
2. Ask one short question: "What is it asking you to find?"
3. Rephrase the instruction in simpler language.
4. Provide a scaffold or limited choice.
5. Model the first step only, then return the task to the pupil.

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TA brief · continued

Planned adaptations

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own [SYS] target on the school's evidence app; the [SYS] carries their EHCP-linked and [SYS] targets, so one record serves every plan.

Follow the pupil's [SYS] where it differs from these routines.

00–03 · Arrival task

At arrival, say the spoken line on the marking card once. Use a truthful last lesson evidence inheritance. Arrival is the only calibration; read the four responses with the starter by minute eight. Choose routes at each task, with no record of switching. IF NEW — 0.5 min read the clues; 1.5 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 1.5 min respond; 1 min sample and reveal.

03–07 · Starter · ready before mixing

Read the arrival to make one reachable adjustment: hold the colour model longer, use the supported We-do prompt, change the current route or give the stretch evidence card. This changes talk within the existing time. IF NEW — 1 min read the two setups; 1 min choose; 2 min explain the lost evidence. IF REVISION — 1 min predict; 2 min justify; 1 min check the method boundary.

07–11 · I do · follow the sampled drops

Model only separated samples going into prepared iodine wells; iodine is not poured into the reaction. Teaching timeline is illustrative, not a practical method or evidence of completion. IF NEW — 2.5 min step through the wells; 1.5 min explain the first observed endpoint. IF REVISION — 0.5 min orient; 2.5 min pupils predict each well meaning; 1 min check the timing window.

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TA brief · continued

11–15 · We do · method-decision checkpoint

Named involvement opportunity. Use card-then-destination by keyboard, touch or pointer; no pointer drag is required. A spill or unidentified solution goes to the adult, not a pupil-designed repair. IF NEW — 1 min read situation and destinations; 2 min place and explain; 1 min respond to a pupil's reasoning. IF REVISION — 0.5 min orient; 2.5 min classify with reasons; 1 min compare the evidence.

15–19 · I do · record what was observed

Use "after 60 s and by 90 s". Do not call 75 s the measured endpoint or claim the true time was known to ±15 s. Sampling uncertainty is not all uncertainty. IF NEW — 2 min model the record; 1 min distinguish observation from inference; 1 min pupil retells. IF REVISION — 1 min inspect; 2 min explain the interval and a control; 1 min compare a missing sample.

19–24 · We do · repair an evidence claim

A pupil can dictate or point to the repair. Keep gaps and ambiguous colours visible; a smooth expected graph is never a reason to change a value. IF NEW — 1 min read the claim; 3 min repair or direct; 1 min check. IF REVISION — 0.5 min read; 3.5 min repair with a reason; 1 min check.

24–34 · You do

Within this twelve-minute block an authorised group may run or record only its assigned trial if the centre method fits; equipment is prepared before class. Use the complete supplied-data route whenever approval, access or time does not fit. Do not run a rushed full pH series or claim paper analysis completes the core practical. IF NEW — 2 min orient the supplied evidence or authorised role; 8 min complete the task; 2 min use one precise feedback point. IF REVISION — 1 min orient; 9 min complete the task; 2 min use one precise feedback point.

34–37 · Review and improve

Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.

37–40 · Exit ticket

Use one book response OR one ticket. Read once to choose the next teaching move for last lesson's rate calculation and evidence claims. No extra receipt, signature, tally or later evidence-chasing. IF NEW — 1 min read evidence; 2 min respond; 1 min pupil finishes the same response. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min pupil finishes the same response.

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TA brief · continued

Planning and evidence

Say once: “I haven't seen what you've been doing… I start from what you show me”.

Lesson code: SCI-LAUNCH-A1-W08-L2. SoW reference: LAUNCH Science, Autumn 1 return block; L2.

Exact SoW outcome: I can follow the centre-approved [AWARD] method to investigate how pH affects amylase activity, recording safe continuous-sampling evidence accurately.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can use continuous-sampling evidence to investigate how pH affects amylase activity.

Evidence (scheme of work): [AWARD] [QUAL] Paper 1 ([QUAL]) topic tests; [AWARD] [QUAL] science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Order the preparation, timing and sampling steps, and justify one evidence decision.; Identify the first observed iodine endpoint and record the time with its sampling limit.; Use matched conditions and raw observations to make a supported comparison.

Evidence location: the pupil’s explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L2. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Staff: confirm the planned adaptations actually used and record any changes below. [SYS] strategies are linked to actual tasks above.

60-second recap of the previous lesson: Amylase is an enzyme; starch is its substrate. Binding depends on the active site, and the enzyme is not used up. The pH plan changes pH, times the first starch-negative sample and keeps other rate factors matched. Read the plan and identify one matched condition.

Must teach core: I can use continuous-sampling evidence to investigate how pH affects amylase activity. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Order the preparation, timing and sampling steps, and justify one evidence decision.

Feedback and evidence

No cold calling; nobody is asked to read difficult text aloud.

Staff mark I, S or C beside this task.